DASHBOARD DESIGN DOCUMENTATION

DePaul University Admissions Interactive Dashboard
Tool Used: Power BI

Dashboard link: https://app.powerbi.com/links/_hnrGnQnwO?ctid=319a61c8-ee1e-4161-8f35-b9553227afd7&pbi_source=linkShare

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1. Purpose of the Dashboard

The purpose of this dashboard is to transform DePaul University's admissions data into a clear, interactive, and insight-driven reporting tool. The dashboard enables stakeholders to monitor application volumes, admission outcomes, applicant demographics, program preferences, and outreach-related activities.

It is designed to:

- Summarize key admissions KPIs at a glance
- Reveal trends and patterns across countries, regions, programs, and intakes
- Support exploratory analysis through interactivity
- Translate descriptive data into actionable insights for admissions strategy and outreach planning

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2. Target Audience

The dashboard is intended for:

- Admissions officers
- Recruitment and outreach teams
- Institutional planning stakeholders
- Data and reporting analysts

The design prioritizes **clarity, usability, and accessibility for non-technical users**.

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3. Data Source and Preparation

Data Source

- Original dataset: Raw DePaul University admissions dataset (Excel)
- Final dataset: Cleaned and analysis-ready Excel dataset loaded into Power BI

Data Preparation Summary

Before visualization, the following steps were completed:

- Removal of duplicate and irrelevant columns
- Standardization of categorical fields (e.g., country, region, degree)
- Identification and documentation of missing values
- Treatment of inconsistent formats
- Validation of date fields
- Retention of high-value fields for admissions and outreach analysis

The cleaned dataset was used directly in Power BI without Power Query transformations, ensuring transparency and traceability.

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4. Key Metrics and KPI Logic

The dashboard includes the following KPIs:

- **Total Applications**
  Count of all application records in the dataset.
- **Total Admitted**
  Calculated using the presence of a non-null *Admit Date* field, due to the absence of a dedicated admission status column.
- **Admission Rate (%)**
  Calculated as:

Total Admitted ÷ Total Applications
Displayed as a percentage.

- **Top Country**
  Country with the highest number of applications (excluding null values).
- **Top Region**
  Region with the highest number of applications (excluding null values).

- **Top Degree**
  Most frequently applied-for degree type.

All KPIs are calculated using DAX measures to ensure accuracy and dynamic filtering.

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5. Dashboard Structure and Page Layout

The dashboard is structured into **multiple pages**, each serving a specific analytical purpose.

Page 1: Admissions Overview

Objective:
Provide a high-level summary of application and admission performance.

Visuals included:

- KPI cards: Total Applications, Total Admitted, Admission Rate
- Line chart: Program start date
- Bar chart: Applications by Country

Interactivity added:

- **Drill-down enabled** on the Applications by Country visual to allow users to explore deeper categorical levels
- **Slicers** for Country, Intake, and Degree to enable dynamic filtering
- **Page navigation buttons** created using formatted blank buttons to allow seamless movement across dashboard pages
- **Tooltips** are used to provide additional context without overcrowding visuals**.**
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Design logic:
This page acts as the executive overview, offering immediate insight into overall admissions performance and geographic trends.

Page 2: Applicant Demographics

Objective:
Analyze the demographic composition of applicants.

Visuals included:

- Distribution of applicants by Country
- Distribution of applicants by Region
- Distribution of applicants by Degree

Design logic:
This page helps identify dominant applicant segments and highlights diversity patterns across geography and academic background.

Page 3: Program and Intake Insights

Objective:
Understand program demand and applicant preferences.

Visuals included:

- Application distribution by Intake
- Total applications by College
- Total applications by College First Choice
- Admission rate by College First Choice

Design logic:
This page supports program-level evaluation by showing which colleges attract the most interest and how admission outcomes vary by applicant preference.

Page 4: Outreach and Engagement Analysis

Objective:
Provide insight into recorded outreach and engagement activities.

Visuals included:

- Count of outreach attempts by counselor name
- Campaign participation by counselor

- Total admitted by date of contact
- Total admission by Slate form filled

Design logic:
This page offers a process-level view of engagement activity. While outreach data is limited, the visuals help identify patterns and highlight data quality constraints for future improvement.

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6. Interactivity Features Implemented

The dashboard includes the following interactive elements:

- **Slicers:** Country, Intake, Degree, College
- **Drill-down:** Enabled on country-level application visual
- **Page navigation:** Implemented using custom-formatted blank buttons
- **Cross-filtering:** Visuals interact dynamically when selections are made
- **Tooltips:** Used to provide additional context without cluttering visuals

These features enhance exploration and user-driven analysis.

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7. Visualization and Design Principles

The dashboard follows best practices for clarity and usability:

- Consistent color scheme across all pages
- Clear, professional titles and labels
- Logical grouping of related visuals
- Minimal color overuse
- Balanced spacing and alignment
- KPIs emphasized only on the overview page to avoid redundancy

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8. Data Flow and Model Design

Data flow:
Raw Excel → Data cleaning → Cleaned Excel → Power BI Data Model → DAX Measures → Visuals

The dashboard uses a single-table model with calculated measures, ensuring simplicity and performance efficiency.

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9. Assumptions and Constraints

- Admission status is inferred from Admit Date
- Outreach metrics are limited due to data values
- Some fields contain missing values and are excluded from top-ranking KPIs
- The analysis focuses on descriptive and exploratory insights rather than predictive modeling

10. Conclusion

This interactive dashboard successfully converts a complex admissions dataset into a structured, user-friendly analytical tool. Through effective visualization, interactivity, and documented assumptions, the dashboard provides actionable insights while acknowledging data limitations. It establishes a strong foundation for future analytical enhancement and improved data collection processes.

INSIGHT REPORT

DePaul University Admissions Dashboard

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1. Overview of Key Findings

This insight report summarizes the major trends, patterns, and observations identified from the interactive admissions dashboard built in Power BI. The analysis focuses on application volume, geographic distribution, program preferences, admission outcomes, and outreach-related activity. Insights are derived from cleaned admissions data and supported by visual evidence from the dashboard.

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2. Application Volume and Admission Outcomes

Insight 1: High Application Volume with Moderate Admission Rate

The dashboard shows a high number of total applications relative to total admissions, resulting in an admission rate of approximately **65%**.

Why this matters:
This suggests a moderately selective admissions process. While a significant proportion of applicants are admitted, there is still a meaningful gap between applications and admissions, indicating room for improving applicant quality, targeting, or follow-up efficiency.

Recommendation:
Admissions teams can analyze applicant segments with higher admission success rates and prioritize similar profiles in future recruitment efforts.

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3. Geographic Distribution of Applicants

Insight 2: Applications Are Concentrated in a Few Key Countries

The country-level analysis reveals that a small number of countries account for the largest share of applications, with one country emerging as the top contributor.

Why this matters:
Heavy dependence on a few countries increases geographic concentration risk. Changes in policy, competition, or economic conditions in those countries could significantly impact application volume.

Recommendation:

- Strengthen recruitment efforts in high-performing countries
- Explore underrepresented regions with growth potential

Insight 3: Regional Trends Mirror Country Concentration

When viewed by region (excluding null values), applications are similarly concentrated in specific regions.

Why this matters:
Regional trends help inform broader international recruitment strategies beyond individual countries.

Recommendation:
Admissions teams can allocate marketing and outreach resources at a regional level rather than country-by-country for efficiency.

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4. Academic and Program Preferences

Insight 4: Degree-Level Demand Is Uneven

The degree distribution shows that **Bachelor's degree applications dominate**, with significantly fewer applications for other degree types.

Why this matters:
This imbalance suggests that undergraduate programs drive the majority of demand.

Recommendation:

- Undergraduate-focused campaigns may yield higher returns
- Graduate programs may require more targeted awareness and positioning

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Insight 5: College and First-Choice Preferences Highlight Program Popularity

Certain colleges and college first-choice selections receive significantly higher application volumes.

Why this matters:
This indicates strong brand appeal or perceived value in specific academic units.

Recommendation:

- Leverage popular colleges as flagship offerings
- Investigate reasons behind low-interest programs and improve visibility or curriculum messaging

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5. Intake Patterns and Timing

Insight 6: Application Distribution Varies Across Intakes

The intake analysis shows uneven application volume across different intake periods.

Why this matters:
Seasonal or intake-based variation affects staffing, processing timelines, and outreach planning.

Recommendation:
Admissions teams can:

- Prepare resources ahead of high-volume intake periods
- Increase marketing activity during historically low-volume intakes

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6. Outreach and Engagement Analysis

Insight 7: Outreach Activity Is Uneven Across Counselors

Counts of outreach attempts and campaigns vary significantly by counselor.

Why this matters:
This suggests differences in workload distribution, engagement strategy, or data capture practices.

Opportunity:

- Standardize outreach tracking
- Share best practices from high-activity counselors
- Improve outreach data completeness for deeper performance analysis

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Insight 8: Admission Outcomes Align with Contact Timing

Admissions linked to dates of contact indicate that applicant engagement timing may influence outcomes.

Why this matters:
Early or timely engagement could increase admission success.

Recommendation:
Admissions teams can:

- Prioritize faster follow-ups
- Track response time as a future KPI

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7. Data Quality Observations Impacting Insights

- Some outreach-related fields contain high proportions of null or binary values, limiting advanced performance analysis
- Admission status had to be inferred from Admit Date due to the absence of a direct admitted field
- Despite these limitations, the cleaned dataset supports strong descriptive and exploratory analysis

These observations highlight areas for improvement in future data collection.

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8. Key Recommendations

Based on the insights derived from the dashboard:

1. **Strengthen recruitment in top-performing countries and regions** while exploring new markets to reduce concentration risk.
2. **Focus undergraduate recruitment strategies**, as bachelor's programs attract the majority of applicants.
3. **Optimize intake-based planning** by aligning outreach and staffing with high-demand periods.
4. **Improve outreach data capture** to enable more accurate measurement of counselor performance and conversion impact.
5. **Introduce standardized admission and engagement flags** (e.g., admitted = Yes/No) to support future predictive and performance analysis.

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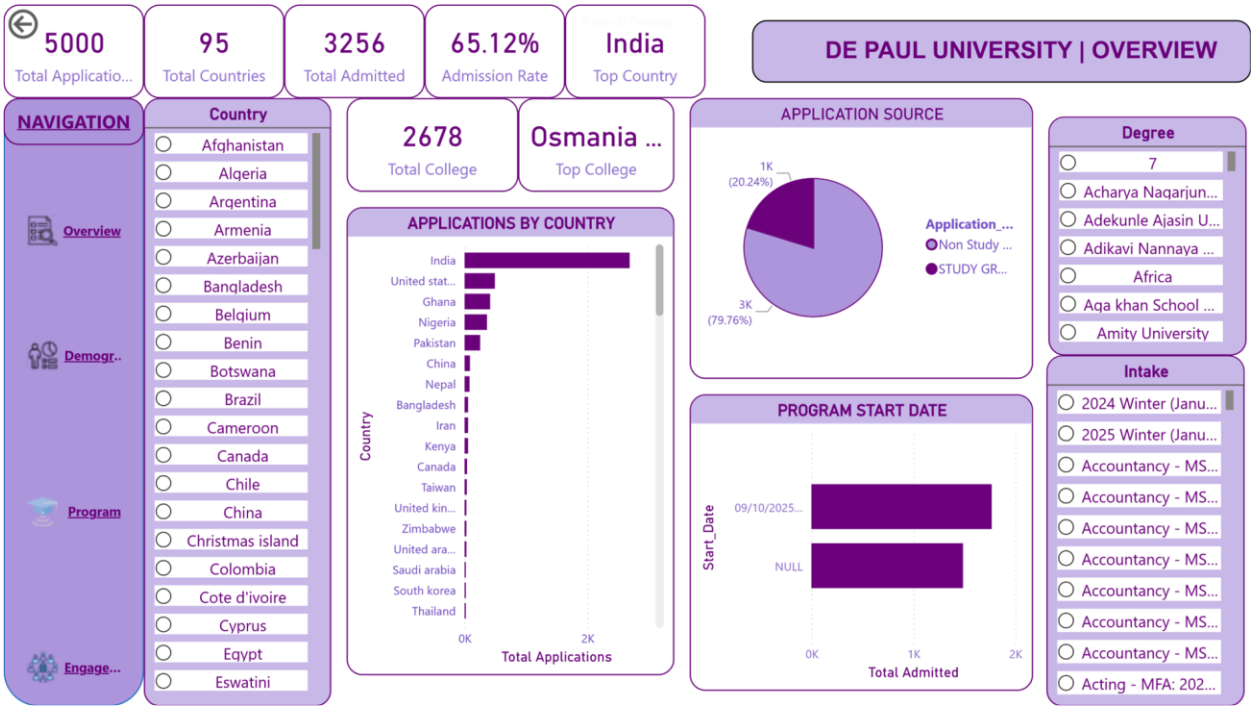
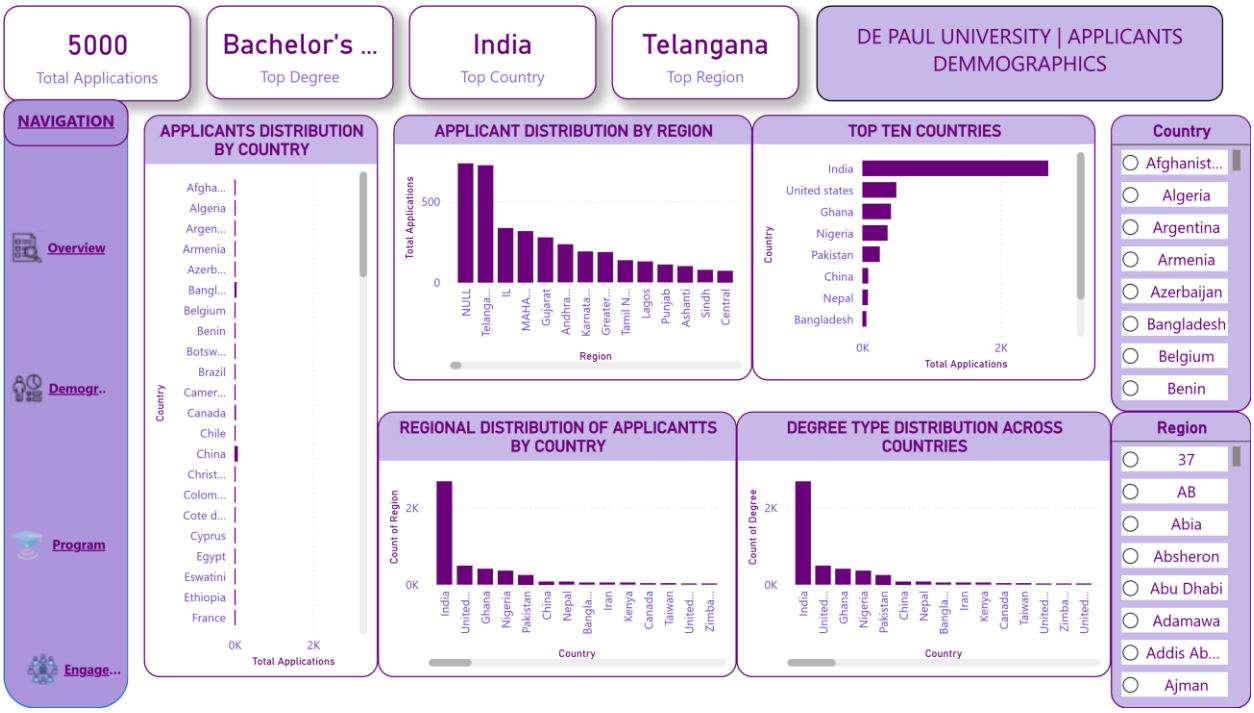
9. Conclusion

The dashboard-driven analysis provides a clear, data-backed view of DePaul University's admissions landscape. Through effective visualization and interactivity, key trends in geography, program demand, intake timing, and engagement have been identified. While data limitations exist, the insights generated offer actionable guidance and establish a strong analytical foundation for future optimization.

Dashboard pages





2678	Osmania ...	3256	65.12%	Jarvis Co...
Total College	Top College	Total Admitted	Admission Rate	Top College 1st c...

DE PAUL UNIVERSITY | PROGRAM AND INTAKE ANALYSIS

NAVIGATION

- Overview
- Demogr..
- Program
- Engage...

College

- ,Chennai,Saveet...
- A.N.U college of ...
- A.R. Junior Colle...
- A.V. College of A...
- Abbottabad univ...
- Abdul Wali Khan...
- Abdul Wali Khan...
- Abia State Unive...

College_1st_Choice

- College of Com...
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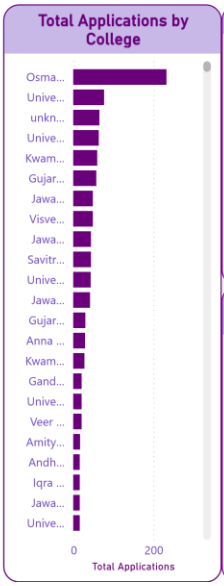
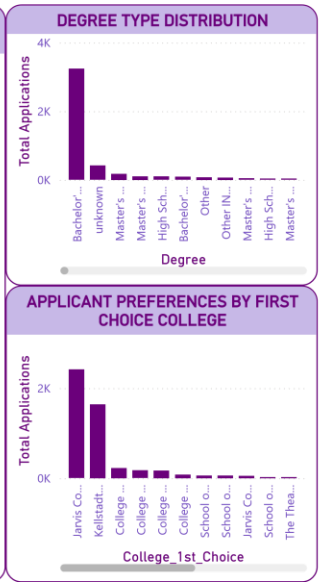
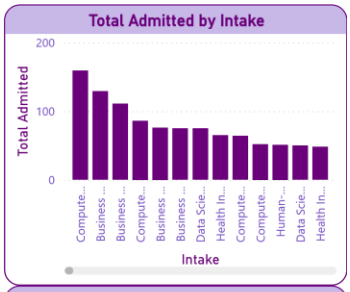
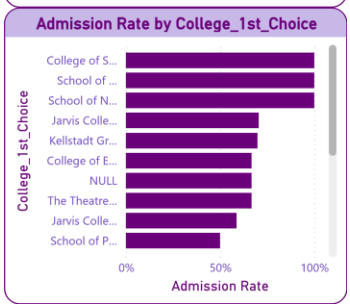
- College of Educa...
- College of Libera...
- College of Scien...
- College of Scien...
- Jarvis College of ...
- Jarvis College of ...









NAVIGATION

- Overview
- Demogr..
- Program
- Engage...

Date_of_Contact	Count of Campaig
01/02/2024 00:00:00	
01/03/2024 00:00:00	
01/04/2024 00:00:00	
01/05/2024 00:00:00	
01/07/2024 00:00:00	
01/08/2023 00:00:00	
01/08/2024 00:00:00	
01/09/2024 00:00:00	
01/10/2024 00:00:00	
01/11/2023 00:00:00	
01/11/2024 00:00:00	
01/12/2023 00:00:00	
01/12/2024 00:00:00	
01/15/2024 00:00:00	
01/16/2024 00:00:00	
01/17/2024 00:00:00	
01/18/2024 00:00:00	
01/19/2024 00:00:00	
01/22/2024 00:00:00	
01/23/2024 00:00:00	
01/24/2024 00:00:00	
01/25/2024 00:00:00	
01/29/2024 00:00:00	
01/30/2024 00:00:00	
01/31/2024 00:00:00	
02/01/2024 00:00:00	
02/02/2024 00:00:00	
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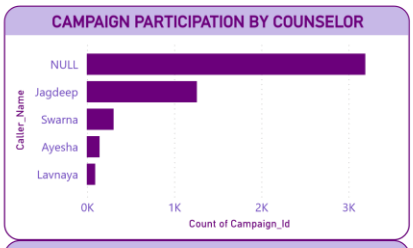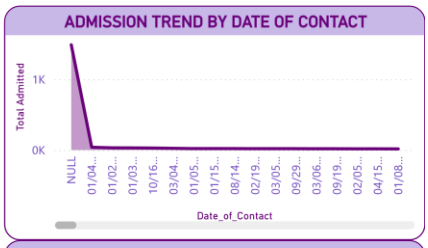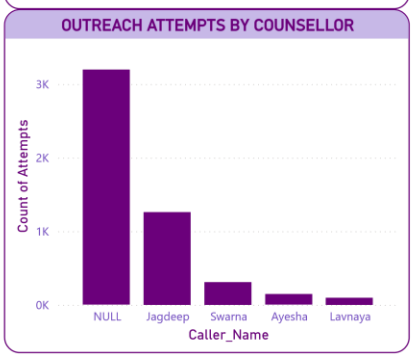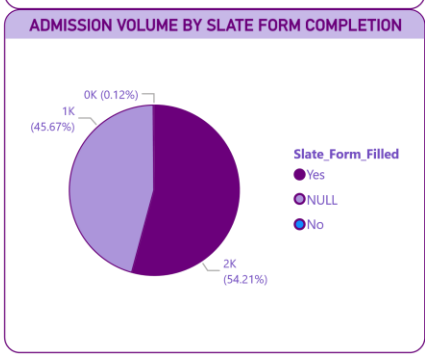
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